



NORTH SYDNEY BOYS HIGH SCHOOL

2023 YEAR 12 HSC ASSESSMENT TASK 3

Mathematics Advanced

General Instructions

- Reading time – 10 minutes
- Working time – **3 hours**
- Write using blue or black pen
- Board approved calculators may be used
- All necessary working should be shown in every question
- Attempt all questions

Class Teacher:

(Please tick or highlight)

- ☐ Ms Sarofim
- ☐ Ms Fu
- ☐ Mr Lin
- ☐ Mr Ireland
- ☐ Dr Vranesevic

Student Number:

(To be used by the exam markers only.)

Question No	1-10	11- 30	Total
Mark	<u>10</u>	<u>90</u>	<u>100</u>

Outcomes – Section I

[illegible]

Section I

10 marks

Allow about 15 minutes for this section. Use the multiple-choice answer sheet

1 The period of the function $f(x) = 2 \tan(4x - \frac{\pi}{3})$ is

- A $\frac{\pi}{2}$
- B $\frac{\pi}{3}$
- C $\frac{\pi}{4}$
- D π

2 For what values of x is the curve $f(x) = x^4 - 2x^3$ concave down?

- A $[0, \frac{3}{2}]$
- B $(0, \frac{3}{2})$
- C $(0, 1)$
- D $[0, 1]$

3 Which expression is the derivative of $\cos^2 3x$ when differentiated with respect to x ?

- A $-6 \sin 3x \cos 3x$
- B $-2 \sin 3x \cos 3x$
- C $2 \sin 3x \cos 3x$
- D $6 \sin 3x \cos 3x$

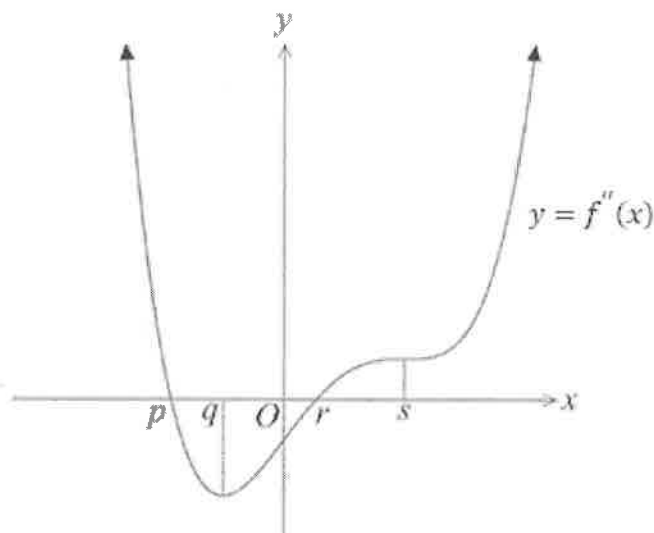
4 The amount M of a drug present in the blood after t hours is given by

$$M = 9t^2 - t^3 \text{ for } 0 \leq t \leq 9.$$

When is the amount of drug in the blood increasing most rapidly?

- A $t = 0$
- B $t = 9$
- C $t = 6$
- D $t = 3$

- 5 The diagram shows the graph of $y = f''(x)$ for the function $f(x)$.



For what value of x does the function $f'(x)$ have a maximum turning point?

- A $x = p$
- B $x = q$
- C $x = r$
- D $x = s$

- 6 The table below shows the probability distribution of a discrete random variable X which has mean -0.45 .

x	-2	-1	0	1	2
$P(X = x)$	0.1	a	0.2	0.15	b

What are the values of a and b ?

- A $a = 0.2, b = 0.35$
- B $a = 0.3, b = 0.25$
- C $a = 0.4, b = 0.15$
- D $a = 0.5, b = 0.05$

- 7 The graph of the function $y = f(x)$ is known to have a minimum turning point at $P(6, -4)$. Therefore the graph of $y = -f(-2x)$ will have a maximum turning point at

- A $(3, 4)$
- B $(-3, 4)$
- C $(-3, -4)$
- D $(-12, 4)$

- 8 Let X and Y be two events such that $P(X) = 0.5$, $P(Y) = 0.6$, and $P(Y|X) = 0.7$.

Which of the following statements is FALSE?

- A $P(X|Y) < P(Y|X)$
- B X and Y are independent events
- C $P(X \cap Y) = 0.35$
- D $P(X \cup Y) = 0.75$

- 9 What are the domain and range of the function $f(x) = \ln(x+1) - \sqrt{4-x^2}$?

- A domain $(-1, \infty)$ range $(-\infty, \infty)$
- B domain $(-1, \infty)$ range $(-\infty, \ln 3]$
- C domain $(-1, 2]$ range $(-\infty, \infty)$
- D domain $(-1, 2]$ range $(-\infty, \ln 3]$

- 10 Which of the following is the correct function value at the minimum turning point of

$$f(x) = (x - 2021)(x - 2022)(x - 2023)$$

- A $\frac{1}{\sqrt{3}}$
- B $-\frac{1}{\sqrt{3}}$
- C $-\frac{2\sqrt{3}}{9}$
- D $\frac{2\sqrt{3}}{9}$

2023 TRIAL HIGHER SCHOOL CERTIFICATE EXAMINATION

Class and Teacher:

Student Number:

Mathematics Advanced Section II

90 marks

Attempt Questions 11-30

Allow about 2 hours and 45 minutes for this section

Instructions

- Write your Student Number at the top of this page.
- Answer the questions in the spaces provided. These spaces provide guidance for the expected length of response.
- Your responses should include relevant mathematical reasoning and/or calculations.
- Extra writing space is provided at the back. If you use this space, clearly indicate which question you are answering.

	2023 Y12 Adv - Assessment Task 3 – Outcomes – Section II											
Question	MA11-3	MA11-4	MA11-6	MA11-7	MA12-1	MA12-3	MA12-4	MA12-5	MA12-6	MA12-7		
11							/3					
12		/2										
13			/3									
14									/3			
15										/7		
16					/3							
17							/4					
18	/5											
19				/5								
20										/4		
21						/8						
22										/5		
23										/5		
24				/5								
25								/6				
26				/4								
27				/2								
28					/4							
29									/5			
30						/7						
Total	/5	/2	/3	/16	/7	/15	/7	/6	/8	/21	/90	

Question 11 (3 marks)

The third term of an arithmetic series is 32 and the sixth term is 17.

- (i) Find the common difference 1

- (ii) Find the sum of the first ten terms. 2

Question 12 (2 marks)

If $\sin \theta = \frac{3}{5}$ and $\tan \theta < 0$, find the exact value of $\cos \theta$ 2

Question 13 (3 marks)

Given that $2\log_e(x^2y) = 3 + \log_e x - \log_e y$, express y in terms of x in simplest terms. **3**

.....

.....

.....

.....

.....

.....

.....

.....

Question 14 (3 marks)

Find the equation of the normal to $y = e^{\cos x}$ at the point where $x = \frac{\pi}{2}$ **3**

.....

.....

.....

.....

.....

.....

.....

.....

.....

Question 15 (7 marks)

(i) Find $\int \sec^2 3x \, dx$

1

.....

.....

(ii) Find $\int \frac{1}{(3x+2)^4} dx$

2

.....

.....

.....

.....

.....

.....

.....

(iii) Evaluate $\int_0^1 e^{-2x} dx$ exactly.

2

.....

.....

.....

.....

.....

.....

.....

(iv) Find $\int \frac{x^2}{x^3-5} dx$

2

Question 16 (3 marks)

The curve $y = \sin x$ is stretched horizontally by a factor of 2, then it is shifted $\frac{\pi}{2}$ units right, then it is stretched vertically by a factor of 3 and reflected in the x -axis.

What equation describes the final curve after this sequence of transformations?

3

Question 17 (4 marks)

A new brand of electric bicycle is introduced to the market and 18,000 are sold in the first month. Each month thereafter, the sales are 70% of the sales in the previous month.

- (i) In which month will monthly sales first drop below 1000 per month? **2**

.....

.....

.....

.....

.....

.....

- (ii) How many bicycles are sold in total in the first year? **1**

.....

.....

.....

.....

- (iii) How many bicycles are eventually sold altogether? **1**

.....

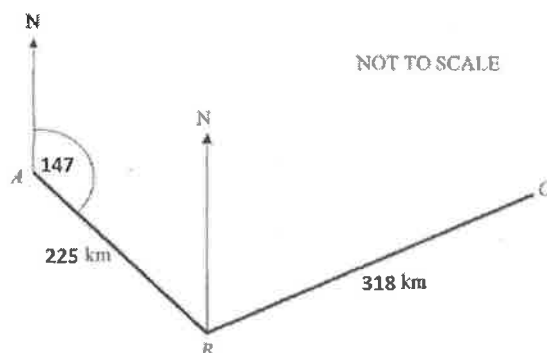
.....

.....

.....

Question 18 (5 marks)

A ship sails 225 km from Adhiban Island on a bearing of 147 degrees and arrive at Port Bologan to pick up some cattle. It then progresses to its destination, Port Cramling, a distance of 318 km on a bearing of 071 degrees.



- (i) Show that $\angle ABC = 104^\circ$
(you may write on the diagram above)

1

- (ii) Show that the distance AC is approximately 431.7 km.

2

.....

.....

.....

.....

.....

.....

- (iii) The return trip is a straight line back to Adhiban Island and not passing through Port Bologan. Find the bearing that the ship must take to go straight from Port Cramling to Adhiban Island.

2

Question 19 (5 marks)

Mischa likes to drink pearl milk tea at work. The number X of teas she drinks each day is a random variable with probability distribution given by:

x	0	1	2	3
$P(X = x)$	0.1	0.2	0.3	0.4

- (i) What is the expected value $E(X)$?

1

- (ii) What are the variance, $\text{Var}(X)$, and the standard deviation, σ ?

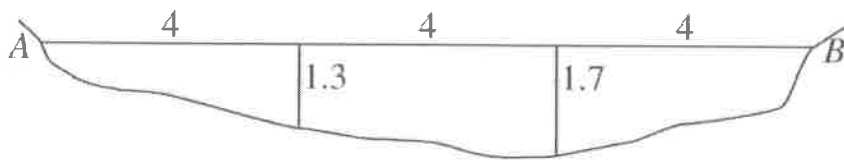
2

- (iii) Mischa is at work on two successive days. What is the probability that she drinks the same number of pearl milk teas on both days?

2

Question 20 (4 marks)

The diagram below shows the cross-section of a stream with the depths of the stream shown in metres at 4 metre intervals. The creek is 12 metres wide.



- (i) Use the trapezoidal rule to approximate the area of the cross-section.

2

- (ii) If water flows through this part of the stream at a speed of 0.5 metres/sec, calculate the approximate volume of water that flows past this section in 1 hour.

2

Question 21 (8 marks)

Consider the function $y = x^3 - 9x^2 + 24x$.

- (i) Find all stationary points and determine their nature.

4

[illegible]

(ii) Find the point of inflection.

2

.....

.....

.....

.....

.....

.....

.....

(iii) Sketch the curve showing all important features.

2

Question 22 (5 marks)

The line L is the tangent to the curve $y = x^3 + 7$ at $x = 2$.

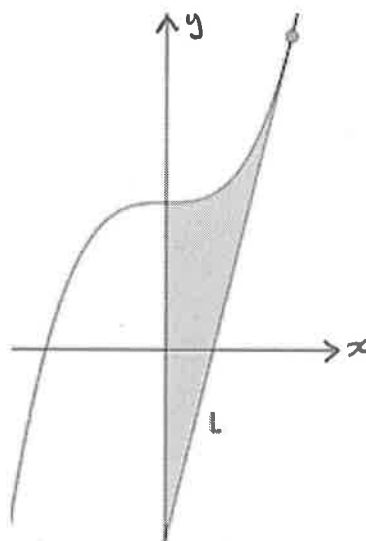
- (i) Show that the equation of the tangent L is $y = 12x - 9$

.....

.....

.....

.....



2

- (ii) Find the area bounded by the y -axis, the tangent L , and the curve $y = x^3 + 7$

3

.....

.....

.....

.....

.....

.....

.....

.....

.....

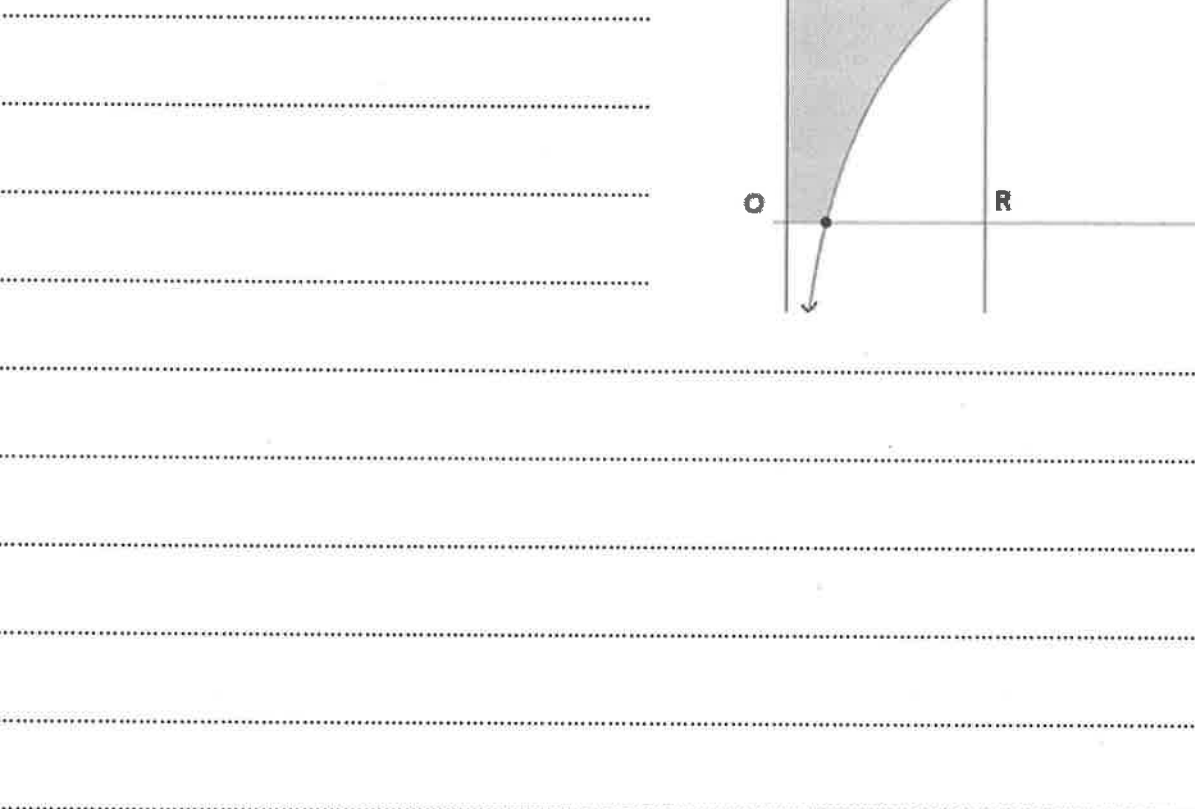
.....

.....

-

-
-
-

- Using your answers from parts (i) and (ii), or otherwise, find the area of the shaded region.



Question 24 (5 marks)

A six-sided die is biased so that the number 5 occurs twice as often as any other number.

- (i) The die is rolled once. Show that the probability that an odd number occurs $\frac{4}{7}$. **1**

- (ii) If the biased die is rolled twice, find the probability that the sum of the uppermost numbers is seven. **2**

The biased die is now rolled together with TWO fair six-sided dice.

(iii) What is the chance that at least two odd numbers are uppermost?

2

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

Question 25 (6 marks)

The outside temperature (in degrees Celsius) on a certain day was modelled

by $T = 12 + 7\sin\left(\frac{\pi t}{12}\right)$ where t is the number of hours after 6am.

- (i) What is the maximum temperature in the day? **1**

- (ii) Sketch a graph of the function $T = 12 + 7\sin\left(\frac{\pi t}{12}\right)$ for $0 \leq t \leq 24$ **2**

- (iii) Between what times during the day is the temperature 15°C or above? **3**

Question 26 (4 marks)

Mrs McCrone walks her three labradoodles at Balmoral Beach every Saturday morning. The dogs are poorly behaved: if a stranger pats them, the chance that the white dog bites him is $\frac{1}{20}$, the chance that the brown one bites him is $\frac{1}{10}$, and the chance that the deranged black one bites him is $\frac{1}{2}$.

- (i) If a stranger selects one of the dogs at random and pats it, what is the chance they will be bitten? **2**

- (ii) Given that a stranger pats one of the dogs and is bitten, what is the probability that it was the black one they patted? **2**

Question 27 (2 marks)

Edward plays a game in which he has a probability p of winning, probability q of losing, and probability r of moving to the next round ($p + q + r = 1$).

What is his probability of eventually winning, in terms of p and q ?

2

This image shows a full page of white paper with horizontal dashed lines, typical of primary school writing paper. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

Question 28 (4 marks)

The point $A(6, 1)$ lies on $h(x)$. The tangent at A is $y = \frac{x}{6}$. Point B is the image of A on the function $g(x) = 3h(2x + 4)$.

(i) Show that B has coordinates $(1, 3)$.

1

[illegible]

(ii) Hence find the equation of the tangent to $g(x)$ at B .

3

This image shows a full page of a handwriting practice worksheet. It consists of multiple sets of three horizontal dashed lines spaced evenly down the page, providing a guide for letter height and placement. The background is plain white, and there are no other markings or text present.

Question 29 (5 marks)

Consider the function $f(x) = \frac{\ln x}{x}$ for $x > 0$.

- (i) Show that the graph of $y = f(x)$ has a stationary point at $x = e$.

2

- (ii) By considering the gradient on either side of $x = e$, or otherwise, show that the stationary point at $x = e$ is a maximum.

1

Question 30 (7 marks)

A truck is making a 1000 kilometre trip at a constant speed of v km/h.

When travelling at v km/h, the truck uses fuel at a rate of $(6 + \frac{v^2}{50})$ litres per hour.

The truck company pays \$2.00 per litre for fuel and pays each of the two drivers \$35 per hour while the truck is travelling.

- (i) Let the total cost of fuel and the driver's pay for the trip be C dollars.

Show that $C = \frac{82000}{v} + 40v$

3

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

- (ii) The truck must take no longer than 12 hours to complete the trip, and speed limits require that $v \leq 110$.

At what speed v should the truck travel to minimise the cost C ?

4

(you may disregard any change-over time for the drivers to swap).

.....

.....

.....

.....

.....

[illegible]

ENDS



NORTH SYDNEY BOYS HIGH SCHOOL

2023 YEAR 12 HSC ASSESSMENT TASK 3

Mathematics Advanced

General Instructions

- Reading time – 10 minutes
- Working time – **3 hours**
- Write using blue or black pen
- Board approved calculators may be used
- All necessary working should be shown in every question
- Attempt all questions

Class Teacher:

(Please tick or highlight)

- ☐ Ms Sarofim
- ☐ Ms Fu
- ☐ Mr Lin
- ☐ Mr Ireland
- ☐ Dr Vranesevic

Student Number:

(To be used by the exam markers only.)

Question No	1-10	11- 30	Total
Mark	<u>10</u>	<u>90</u>	<u>100</u>

Outcomes – Section I

[illegible]

Section I

1	2	3	4	5	6	7	8	9	10
C	C	A	D	A	D	B	B	D	C

Allow about 15 minutes for this section. Use the multiple-choice answer sheet

- 1 The period of the function $f(x) = 2 \tan(4x - \frac{\pi}{3})$ is

A $\frac{\pi}{2}$

B $\frac{\pi}{3}$

C $\frac{\pi}{4}$ ✓

D π

- 2 For what values of x is the curve $f(x) = x^4 - 2x^3$ concave down?

A $[0, \frac{3}{2}]$

B $(0, \frac{3}{2})$

C $(0, 1)$ ✓

D $[0, 1]$

- 3 Which expression is the derivative of $\cos^2 3x$ when differentiated with respect to x ?

A $-6 \sin 3x \cos 3x$ ✓

B $-2 \sin 3x \cos 3x$

C $2 \sin 3x \cos 3x$

D $6 \sin 3x \cos 3x$

- 4 The amount M of a drug present in the blood after t hours is given by

$$M = 9t^2 - t^3 \text{ for } 0 \leq t \leq 9.$$

When is the amount of drug in the blood increasing most rapidly?

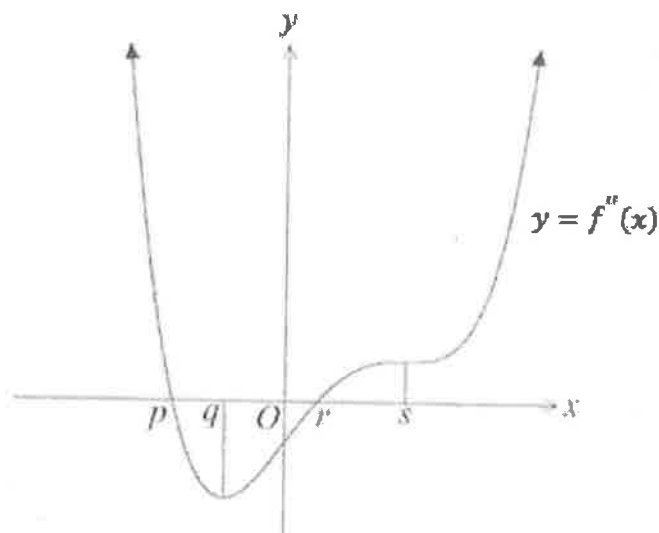
A $t = 0$

B $t = 9$

C $t = 6$

D $t = 3$ ✓

- 5 The diagram shows the graph of $y = f''(x)$ for the function $f(x)$.



For what value of x does the function $f'(x)$ have a maximum turning point?

- A $x = p$ ✓
- B $x = q$
- C $x = r$
- D $x = s$

- 6 The table below shows the probability distribution of a discrete random variable X which has mean -0.45 .

x	-2	-1	0	1	2
$P(X=x)$	0.1	a	0.2	0.15	b

What are the values of a and b ?

- A $a = 0.2, b = 0.35$
- B $a = 0.3, b = 0.25$
- C $a = 0.4, b = 0.15$
- D $a = 0.5, b = 0.05$ ✓

- 7 The graph of the function $y = f(x)$ is known to have a minimum turning point at $P(6, -4)$. Therefore the graph of $y = -f(-2x)$ will have a maximum turning point at

- A $(3, 4)$
- B $(-3, 4)$ ✓
- C $(-3, -4)$
- D $(-12, 4)$

- 8 Let X and Y be two events such that $P(X) = 0.5$, $P(Y) = 0.6$, and $P(Y|X) = 0.7$.

Which of the following statements is FALSE?

- A $P(X|Y) < P(Y|X)$
- B X and Y are independent events ✓
- C $P(X \cap Y) = 0.35$
- D $P(X \cup Y) = 0.75$

- 9 What are the domain and range of the function $f(x) = \ln(x+1) - \sqrt{4-x^2}$?

- A domain $(-1, \infty)$ range $(-\infty, \infty)$
- B domain $(-1, \infty)$ range $(-\infty, \ln 3]$
- C domain $(-1, 2]$ range $(-\infty, \infty)$
- D domain $(-1, 2]$ range $(-\infty, \ln 3]$ ✓

- 10 Which of the following is the correct function value at the minimum turning point of $f(x) = (x-2021)(x-2022)(x-2023)$

- A $\frac{1}{\sqrt{3}}$
- B $-\frac{1}{\sqrt{3}}$
- C $-\frac{2\sqrt{3}}{9}$ ✓
- D $\frac{2\sqrt{3}}{9}$

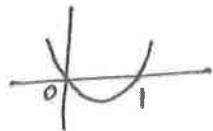
2023 - Y12 Advanced Task 3
Multiple Choice

① $f(x) = 2 \tan(4x - \frac{\pi}{3})$

$\therefore \text{period} = \frac{\pi}{4} \therefore \text{C}$

② $f(x) = x^4 - 2x^3 \therefore f'(x) = 4x^3 - 6x^2$

$\therefore f''(x) = 12x^2 - 12x = 12x(x-1)$



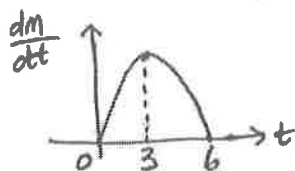
$\therefore 0 < x < 1 \therefore \text{C}$

③ $y = (\cos 3x)^2 \therefore y' = 2 \cos 3x \cdot -3 \sin 3x$

$= -6 \sin 3x \cos 3x \therefore \text{A}$

④ $M = 9t^2 - t^3 \therefore \frac{dM}{dt} = 18t - 3t^2$

$= 3t(6-t)$



$\therefore \text{at } t=3 \therefore \text{D}$

⑤ Graph is $y = f''(x)$. For $f'(x)$ to have max. turning point, we want $f''(x) = 0$ and f'' changing from + to - $\therefore x=p \therefore \text{A}$

⑥ Probs. add to 1, $\therefore a+b = 0.55 \dots (1)$

$E(x) = -0.45 \therefore -0.2 - a + 0 + 0.15 + 2b = -0.45$

i.e. $-a + 2b = -0.40 \dots (2)$

① + ② $\rightarrow 3b = 0.15$

$\therefore b = 0.05$

$a = 0.5$

$\therefore \text{D}$

- ⑦ $y = f(x)$
Curve has been
- compressed by factor of 2 $\rightarrow f(2x)$
 - reflected across y axis $\rightarrow f(-2x)$
 - reflected across x axis $\rightarrow -f(-2x)$
- $\therefore (6, -4) \rightarrow (3, -4) \rightarrow (-3, -4) \rightarrow (-3, 4) \therefore \textcircled{B}$

- ⑧ $P(Y|X) = \frac{P(Y \cap X)}{P(X)} \therefore 0.7 = \frac{P(Y \cap X)}{0.5}$
 $\therefore P(Y \cap X) = 0.35$
- But $P(Y) \cdot P(X) = (0.6)(0.5) = 0.30 \neq P(Y \cap X)$
 $\therefore$ not independent $\therefore \textcircled{B}$

Note: $P(X|Y) = \frac{P(X \cap Y)}{P(Y)} = \frac{0.35}{0.6} \approx 0.58 < P(Y|X) \checkmark$
 $P(X \cap Y) = 0.35 \checkmark$
 $P(X \cup Y) = P(X) + P(Y) - P(X \cap Y)$
 $= 0.5 + 0.6 - 0.35 = 0.75 \checkmark$

- ⑨ $f(x) = \ln(x+1) - \sqrt{4-x^2}$
- $D: (-1, \infty)$ $R: (-\infty, \infty)$
- $D: [-2, 2] \rightarrow D: (-1, 2]$ $\therefore \textcircled{D}$
 $R: [0, 2] \rightarrow R: (-\infty, \ln 3]$

- ⑩ $f(x) = (x-2021)(x-2022)(x-2023)$ is just a horizontal translation of $f(x) = (x+1)(x)(x-1)$, & so will have same y values at min. turning point. Let $f(x) = x(x^2-1) = x^3-x$
 $f'(x) = 3x^2-1 = 0$ when $x = -\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}$.
 $f''(x) = 6x > 0$ when $x = \frac{1}{\sqrt{3}}$
 $\therefore y = \left(\frac{1}{\sqrt{3}}+1\right)\left(\frac{1}{\sqrt{3}}\right)\left(\frac{1}{\sqrt{3}}-1\right) = \frac{1}{\sqrt{3}}\left(\frac{1}{3}-1\right)$
 $= \frac{-2}{3\sqrt{3}} = -\frac{2\sqrt{3}}{9} \therefore \textcircled{C}$

2023 TRIAL HIGHER SCHOOL CERTIFICATE EXAMINATION

Class and Teacher:

Student Number:

Mathematics Advanced Section II

90 marks

Attempt Questions 11-30

Allow about 2 hours and 45 minutes for this section

Instructions

- Write your Student Number at the top of this page.
- Answer the questions in the spaces provided. These spaces provide guidance for the expected length of response.
- Your responses should include relevant mathematical reasoning and/or calculations.
- Extra writing space is provided at the back. If you use this space, clearly indicate which question you are answering.

2023 Y12 Adv - Assessment Task 3 – Outcomes – Section II											
Question	MA11-3	MA11-4	MA11-6	MA11-7	MA12-1	MA12-3	MA12-4	MA12-5	MA12-6	MA12-7	
11							/3				
12		/2									
13			/3								
14									/3		
15										/7	
16					/3						
17							/4				
18	/5										
19				/5							
20										/4	
21						/8					
22										/5	
23										/5	
24				/5							
25								/6			
26				/4							
27				/2							
28					/4						
29									/5		
30						/7					
Total	/5	/2	/3	/16	/7	/15	/7	/6	/8	/21	/90

Question 11 (3 marks)

The third term of an arithmetic series is 32 and the sixth term is 17.

- (i) Find the common difference

1

$$T_3 = a + 2d = 32 \quad \text{--- (1)}$$

$$T_6 = a + 5d = 17 \quad \text{--- (2)}$$

$$\textcircled{1} - \textcircled{2} \rightarrow -3d = 15 \quad \therefore \textcircled{d = -5} \quad \checkmark$$

- (ii) Find the sum of the first ten terms.

2

$$\text{From (i), } a = 32 - 2(-5) \quad \therefore \textcircled{a = 42}$$

$$\begin{aligned} \therefore S_{10} &= \frac{10}{2} (2(42) + (10-1)(-5)) \\ &= 5 (84 - 45) \end{aligned}$$

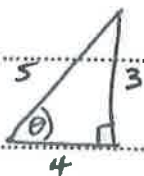
$$\therefore \textcircled{S_{10} = 195} \quad \checkmark \checkmark$$

Question 12 (2 marks)

If $\sin \theta = \frac{3}{5}$ and $\tan \theta < 0$, find the exact value of $\cos \theta$

2

$$\sin > 0, \tan < 0 \quad \therefore \text{Q2} \quad \checkmark \quad \therefore \cos < 0$$



$$\therefore \textcircled{\cos \theta = -\frac{4}{5}} \quad \checkmark \checkmark$$

Question 13 (3 marks)

Given that $2\log_e(x^2y) = 3 + \log_e x - \log_e y$, express y in terms of x in simplest terms. 3

$2\ln(x^2y) = 3 + \ln x - \ln y$ $\therefore 2\ln x^2 + 2\ln y = 3 + \ln x - \ln y$ $4\ln x + 2\ln y = 3 + \ln x - \ln y$ $3\ln x + 3\ln y = 3$ $\ln x + \ln y = 1$ $\ln(xy) = 1$ $\therefore xy = e$ $\therefore y = \frac{e}{x} \quad \checkmark$	<u>ALT:</u> $\ln x^4 y^2 = 3 + \ln x - \ln y$ $\ln x^4 y^2 - \ln x + \ln y = 3$ $\ln \frac{x^4 y^2 \times y}{x} = 3$ $\ln(x^3 y^3) = 3$ $3\ln xy = 3$ $\ln xy = 1$ $xy = e$ $y = \frac{e}{x}$
---	--

Correctly uses log laws $\checkmark\checkmark$

Question 14 (3 marks)

Find the equation of the normal to $y = e^{\cos x}$ at the point where $x = \frac{\pi}{2}$ 3

$$y' = -\sin x \cdot e^{\cos x} \quad \checkmark$$

When $x = \frac{\pi}{2}$, $y = e^{\cos \frac{\pi}{2}} = e^0 = 1$

$$\& \quad y' = -\sin \frac{\pi}{2} \cdot e^{\cos \frac{\pi}{2}} = -1$$

$$\therefore m_n = 1 \quad \checkmark$$

So normal is $y - 1 = 1(x - \frac{\pi}{2})$

$$\therefore y = x + 1 - \frac{\pi}{2} \quad \checkmark$$

(ie $x - y + 1 - \frac{\pi}{2} = 0$)

Question 15 (7 marks)

(i) Find $\int \sec^2 3x \, dx$

1

$$= \frac{1}{3} \tan 3x + C \quad \checkmark$$

(Note: maximum penalty 1 mark for lack of constant)

(ii) Find $\int \frac{1}{(3x+2)^4} \, dx$

2

$$\int (3x+2)^{-4} \, dx = \frac{(3x+2)^{-3}}{-3 \times 3} + C$$

$$\therefore = \frac{-1}{9(3x+2)^3} + C \quad \checkmark \checkmark$$

(iii) Evaluate $\int_0^1 e^{-2x} \, dx$ exactly.

2

$$\int_0^1 e^{-2x} \, dx = \left[-\frac{e^{-2x}}{2} \right]_0^1 \quad \checkmark$$

$$= -\frac{e^{-2}}{2} - \left(-\frac{1}{2} \right)$$

$$\therefore = \frac{1}{2} - \frac{1}{2e^2} \quad \checkmark$$

(ie $\frac{e^2 - 1}{2e^2}$)

(iv) Find $\int \frac{x^2}{x^3-5} dx$

2

$$= \frac{1}{3} \int \frac{3x^2}{x^3-5} dx$$

$$= \frac{1}{3} \ln |x^3-5| + C$$

✓ correct answer

✓ absolute value

Question 16 (3 marks)

The curve $y = \sin x$ is stretched horizontally by a factor of 2, then it is shifted $\frac{\pi}{2}$ units right, then it is stretched vertically by a factor of 3 and reflected in the x-axis.

What equation describes the final curve after this sequence of transformations?

3

$$y = \sin x \rightarrow y = \sin \frac{x}{2}$$

✓ horiz. stretch.

$$\rightarrow y = \sin \frac{1}{2} \left(x - \frac{\pi}{2} \right)$$

✓ shift R

$$\rightarrow y = -3 \sin \frac{1}{2} \left(x - \frac{\pi}{2} \right)$$

✓ vert. stretch & reflection

$$\text{(ie } y = -3 \sin \left(\frac{x}{2} - \frac{\pi}{4} \right) \text{)}$$

Question 17 (4 marks)

A new brand of electric bicycle is introduced to the market and 18,000 are sold in the first month. Each month thereafter, the sales are 70% of the sales in the previous month.

- (i) In which month will monthly sales first drop below 1000 per month?

2

We want $T_n = 18000 \times (0.7)^{n-1} < 1000$

$$\therefore 0.7^{n-1} < \frac{1}{18}$$

$$(n-1) \log_{10} 0.7 < \log_{10} \frac{1}{18}$$

$$n-1 > \frac{\log_{10} \frac{1}{18}}{\log_{10} 0.7} \quad \therefore n > 8.10365 + 1$$

$$n > 9.1$$

$\therefore$ in the 10th month.

- (ii) How many bicycles are sold in total in the first year?

1

$$S_{12} = \frac{18000(1 - 0.07^{12})}{1 - 0.07} \approx 59169.52 \dots$$

$\therefore$ 59169 sold in 1st year

- (iii) How many bicycles are eventually sold altogether?

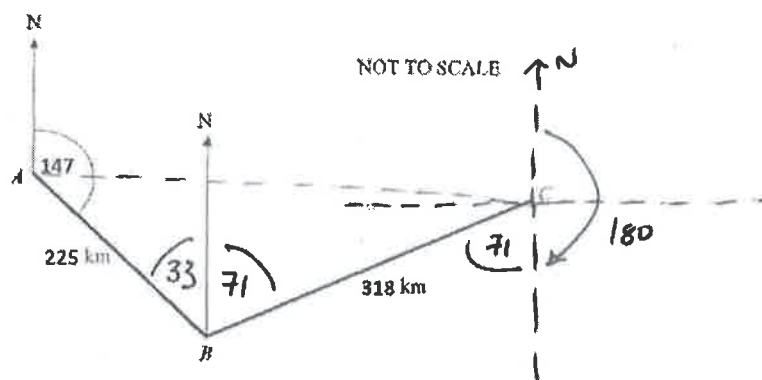
1

as $|r| < 1$ $\therefore$ limiting sum exists

$$\therefore S_{\infty} = \frac{18000}{1 - 0.07} = 60000$$

Question 18 (5 marks)

A ship sails 225 km from Adhiban Island on a bearing of 147 degrees and arrive at Port Bologan to pick up some cattle. It then progresses to its destination, Port Cramling, a distance of 318 km on a bearing of 071 degrees.



- (i) Show that $\angle ABC = 104^\circ$
(you may write on the diagram above)

1

$$\angle ABC = (180^\circ - 147^\circ) + 71^\circ = 33^\circ + 71^\circ = 104^\circ$$

↑
alternate
angle in
parallel
lines

↑
bearing of
C from B

- (ii) Show that the distance AC is approximately 431.7 km.

2

$$AC^2 = 225^2 + 318^2 - 2(225)(318) \cos 104^\circ$$

$$\div 186\,368.0233 \dots$$

$$\therefore AC \div 431.7036 \dots$$

Thus $AC = 431.7 \text{ km (1 d.p.)}$

- (iii) The return trip is a straight line back to Adhiban Island and not passing through Port Bologan. Find the bearing that the ship must take to go straight from Port Cramling to Adhiban Island.

2

$$\frac{\sin \angle ACB}{225} = \frac{\sin 10^\circ}{431.7036}$$

$$\therefore \sin \angle ACB \doteq 0.505709296 \dots$$

$$\therefore \angle ACB \doteq 30^\circ 23'$$

$$\therefore \text{Bearing} = 180^\circ + 71^\circ + 30^\circ 23'$$

$$= \boxed{281^\circ 23' \text{ T.}}$$



Question 19 (5 marks)

Mischa likes to drink pearl milk tea at work. The number X of teas she drinks each day is a random variable with probability distribution given by:

x	0	1	2	3
$P(X=x)$	0.1	0.2	0.3	0.4

- (i) What is the expected value $E(X)$?

1

$$E(X) = 0(0.1) + 1(0.2) + 2(0.3) + 3(0.4)$$

$$= 0 + 0.2 + 0.6 + 1.2 = \boxed{2}$$



- (ii) What are the variance, $\text{Var}(X)$, and the standard deviation, σ ?

2

$$\text{Var}(X) = E(X^2) - \mu^2$$

$$= (0.1)(0^2) + 0.2(1^2) + 0.3(2^2) + 0.4(3^2) - 2^2$$

$$= 5 - 4$$



$$\therefore \boxed{\text{Var}(X) = 1, \sigma = \sqrt{1} = 1}$$



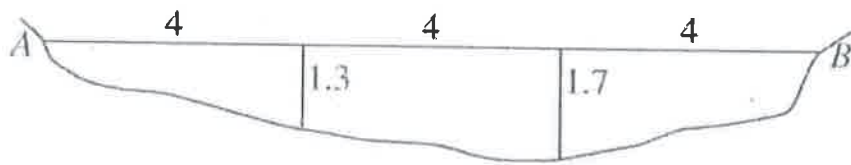
- (iii) Mischa is at work on two successive days. What is the probability that she drinks the same number of pearl milk teas on both days?

2

$$\begin{aligned}
 \text{probability} &= P(0,0) + P(1,1) + P(2,2) + P(3,3) \\
 &= (0.1)^2 + (0.2)^2 + (0.3)^2 + (0.4)^2 \\
 &= 0.01 + 0.04 + 0.09 + 0.16 \\
 \therefore \text{probability} &= 0.3
 \end{aligned}$$

Question 20 (4 marks)

The diagram below shows the cross-section of a stream with the depths of the stream shown in metres at 4 metre intervals. The creek is 12 metres wide.



- (i) Use the trapezoidal rule to approximate the area of the cross-section.

2

x	0	4	8	12
y	0	1.3	1.7	0

$$A \doteq \frac{12-0}{2 \times 3} [0 + 0 + 2(1.3 + 1.7)] = 12 \text{ m}^2$$

- (ii) If water flows through this part of the stream at a speed of 0.5 metres/sec, calculate the approximate volume of water that flows past this section in 1 hour.

2

$$V = A \times \text{length}$$

$$\text{length} = \text{speed} \times \text{time}$$

$$= 0.5 \times (60 \times 60) \text{ m}$$

$$= 1800 \text{ m.}$$

$$\therefore V \doteq 12 \times 1800 \text{ m}^3$$

$$V \doteq 21600 \text{ m}^3$$

$$(\doteq 21600 \text{ KL})$$

Question 21 (8 marks)

Consider the function $y = x^3 - 9x^2 + 24x$.

- (i) Find all stationary points and determine their nature.

4

$$y = x^3 - 9x^2 + 24x$$

$$\therefore y' = 3x^2 - 18x + 24$$

$$= 3(x^2 - 6x + 8)$$

$$y' = 3(x-4)(x-2)$$

For stationary points, $y' = 0 \quad \therefore (x-4)(x-2) = 0$

$$\therefore x = 4$$

$$y = 20$$

or

$$x = 2$$

$$y = 16$$

$$x = 4, x = 2$$

$$y'' = 6x - 18 = 6(x-3)$$

At $(4, 20)$, $y'' = 6(4-3) = 6 > 0$

✓ for testing.

$\therefore$ minimum turning point at $(4, 20)$ ✓
min. point

At $(2, 16)$, $y'' = 6(2-3) = -6 < 0$

$\therefore$ maximum turning point at $(2, 16)$ ✓
max. point.

(ii) Find the point of inflection.

2

For inflections, $y'' = 0 \quad \therefore 6/(x-3) = 0$

$\therefore x = 3, y = 18$ ✓ point

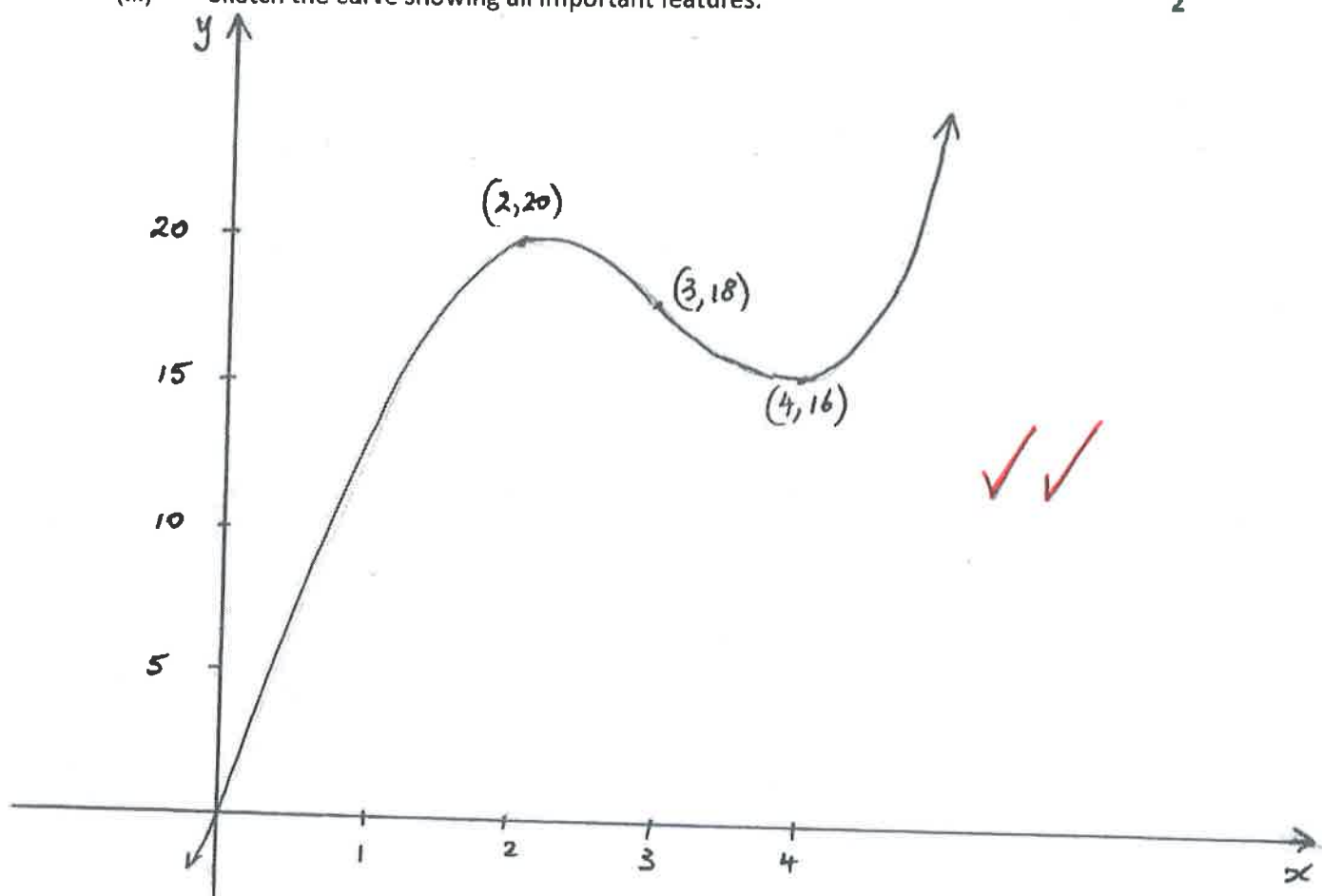
x	2	3	4
y''	-6	0	6

✓ testing.
 $\therefore$ change in concavity at $x = 3$

$\therefore (3, 18)$ is point of inflection.

(iii) Sketch the curve showing all important features.

2



Question 22 (5 marks)

The line L is the tangent to the curve $y = x^3 + 7$ at $x = 2$.

- (i) Show that the equation of the tangent L is $y = 12x - 9$

$$y' = 3x^2$$

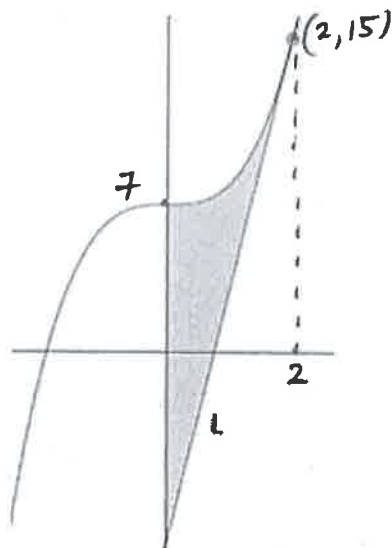
$$\therefore \text{At } x=2, y' = 12 \text{ and } y = 15 \quad \checkmark$$

$\therefore$ tangent equation is

$$y - 15 = 12(x - 2) \quad \checkmark$$

$$y - 15 = 12x - 24$$

$$\therefore y = 12x - 9, \text{ as required.}$$



- (ii) Find the area bounded by the y-axis, the tangent L, and the curve $y = x^3 + 7$

$$A = \int_0^2 [x^3 + 7 - (12x - 9)] dx \quad \checkmark$$

$$= \int_0^2 (x^3 - 12x + 16) dx$$

$$= \left[\frac{x^4}{4} - 6x^2 + 16x \right]_0^2 \quad \checkmark$$

$$= \left[\frac{16}{4} - 6(4) + 16(2) \right] - [0 - 0 + 0]$$

$$\therefore A = 12 \text{ u}^2 \quad \checkmark$$

Question 23 (5 marks)

(i) Show that $\frac{d}{dx}(x \ln x - x) = \ln x$

1

$$\frac{d}{dx}[x \ln x - x] = \ln x \cdot 1 + x \cdot \frac{1}{x} - 1 = \ln x + 1 - 1 = \ln x, \text{ as required.}$$

(ii) Hence or otherwise find $\int \ln x^2 dx$

1

$$\ln x^2 = 2 \ln x$$

$$\therefore \int \ln x^2 dx = 2(x \ln x - x) + C, \text{ from (i).}$$

(iii) The graph shows the curve $y = \ln x^2$, ($x > 0$) which meets the line $x = 5$ at Q. 3

Using your answers from parts (i) and (ii), or otherwise,

find the area of the shaded region.

Area of rectangle OPQR

$$= 5 \times \ln 25$$

$$= 5 \times \ln 5^2$$

$$\therefore \text{Area OPQR} = 10 \ln 5 \text{ u}^2.$$

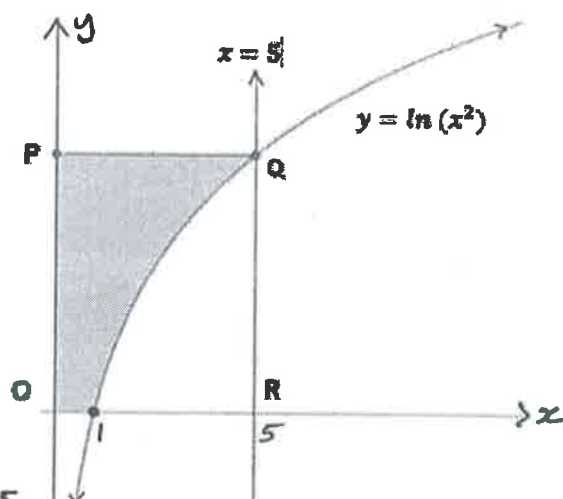
$$\therefore \text{Area shaded region} = 10 \ln 5 - \int_1^5 \ln x^2 dx$$

$$\text{Thus Area} = 10 \ln 5 - 2 \left[x \ln x - x \right]_1^5$$

$$= 10 \ln 5 - 2 \left[(5 \ln 5 - 5) - (\ln 1 - 1) \right]$$

$$= 10 \ln 5 - (10 \ln 5 - 8)$$

$$\therefore \text{Area} = 8 \text{ u}^2$$



Question 24 (5 marks)

A six-sided die is biased so that the number 5 occurs twice as often as any other number.

- (i) The die is rolled once. Show that the probability that an odd number occurs $\frac{4}{7}$.

1

Sample space is $\{1, 2, 3, 4, 5, 5, 6\}$
(these elements are equally likely: we give two 5's since it is twice as likely as the others)

Event is $\{1, 3, 5, 5\}$
 $\therefore P(\text{odd}) = \frac{4}{7}$ ✓

- (ii) If the biased die is rolled twice, find the probability that the sum of the uppermost numbers is seven.

2

$E = \{(2, 5), (5, 2), (3, 4), (4, 3), (1, 6), (6, 1)\}$ ✓
 $\therefore P(E) = \frac{1}{7} \times \frac{2}{7} + \frac{2}{7} \times \frac{1}{7} + 4 \times \left(\frac{1}{7} \times \frac{1}{7}\right)$
 $= \frac{8}{49}$ ✓

The biased die is now rolled together with TWO fair six-sided dice.

(iii) What is the chance that at least two odd numbers are uppermost?

2

"At least 2 odd numbers" means

"3 odd numbers" or "2 odd and 1 even".

Now,

$$P(\text{odd number on fair die}) = \frac{1}{2}, \quad P(\text{odd n}^\circ \text{ on biased die}) = \frac{4}{7}$$

$$P(\text{even number on fair die}) = \frac{1}{2}, \quad P(\text{even n}^\circ \text{ on biased die}) = \frac{3}{7}$$

$$\therefore P(E) = \frac{1}{2} \times \frac{1}{2} \times \frac{4}{7} + P[(O, O, E), \text{ or } (O, E, O) \text{ or } (E, O, O)]$$

biased die biased die

$$= \frac{1}{7} + \frac{4}{7} \times \frac{1}{2} \times \frac{1}{2} + \frac{4}{7} \times \frac{1}{2} \times \frac{1}{2} + \frac{3}{7} \times \frac{1}{2} \times \frac{1}{2}$$

$$= \frac{1}{7} + \frac{11}{28}$$

$$P(E) = \frac{15}{28}$$



Question 25 (6 marks)

The outside temperature (in degrees Celsius) on a certain day was modelled

by $T = 12 + 7\sin\left(\frac{\pi t}{12}\right)$ where t is the number of hours after 6am.

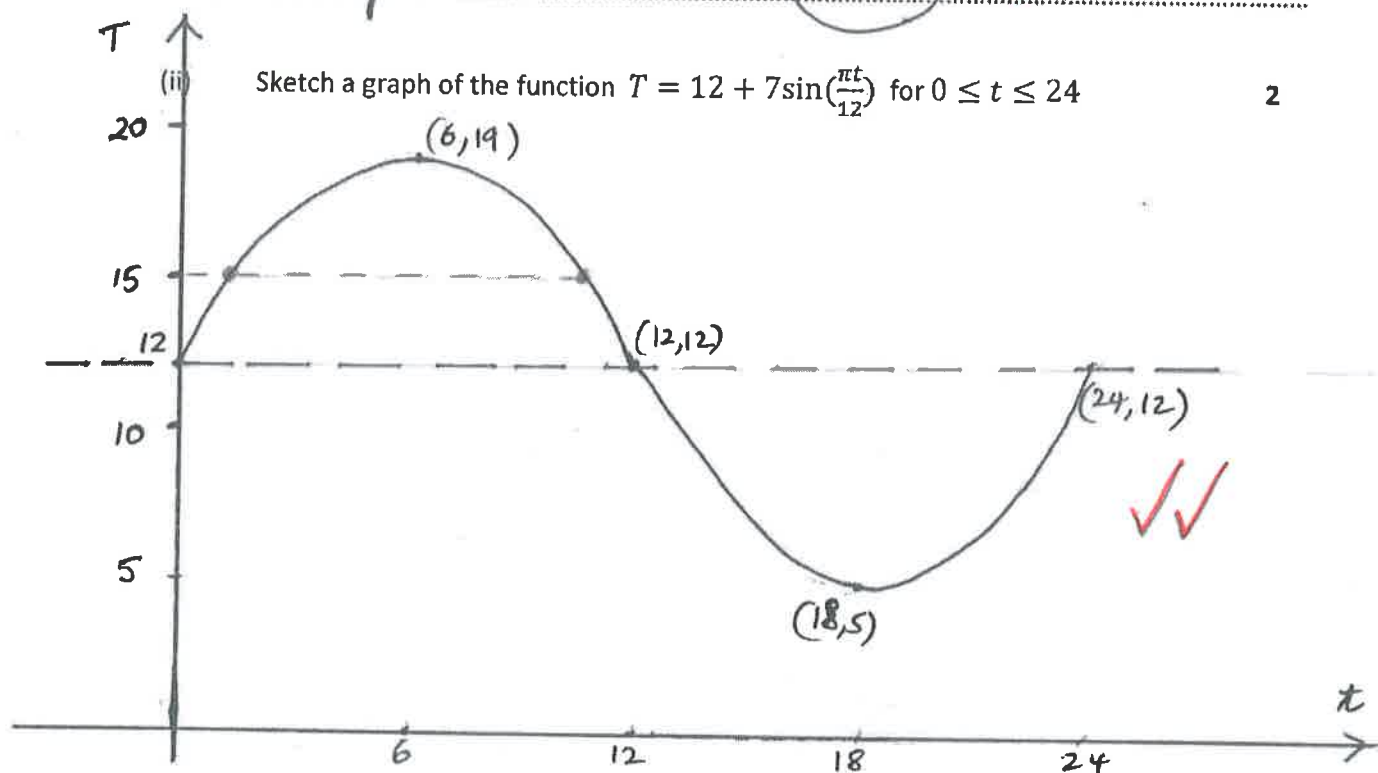
- (i) What is the maximum temperature in the day?

1

Max. temp. = $12 + 7 = 19^\circ\text{C}$ ✓

- (ii) Sketch a graph of the function $T = 12 + 7\sin\left(\frac{\pi t}{12}\right)$ for $0 \leq t \leq 24$

2



- (iii) Between what times during the day is the temperature 15°C or above?

3

Let $12 + 7\sin\left(\frac{\pi t}{12}\right) = 15$

$\therefore \sin\left(\frac{\pi t}{12}\right) = \frac{3}{7}$

Thus $t = \frac{12}{\pi} \sin^{-1} \frac{3}{7}$ or $t = \frac{12}{\pi} \left(\pi - \sin^{-1} \frac{3}{7}\right)$ ✓

$\therefore t \doteq 1.692$ hours or 10.308 hours ✓

ie times are 7.692 or 16.308

$\therefore \geq 15^\circ\text{C}$ from 7:42am and 16:18 pm. ✓

Question 26 (4 marks)

Mrs McCrone walks her three labradoodles at Balmoral Beach every Saturday morning. The dogs are poorly behaved: if a stranger pats them, the chance that the white dog bites him is $\frac{1}{20}$, the chance that the brown one bites him is $\frac{1}{10}$, and the chance that the deranged black one bites him is $\frac{1}{2}$.

- (i) If a stranger selects one of the dogs at random and pats it, what is the chance they will be bitten? 2

$$P(\text{bitten}) = \frac{1}{3} \times \frac{1}{20} + \frac{1}{3} \times \frac{1}{10} + \frac{1}{3} \times \frac{1}{2} \quad \checkmark$$
$$= \frac{13}{60} \quad \checkmark$$

- (ii) Given that a stranger pats one of the dogs and is bitten, what is the probability that it was the black one they patted? 2

$$P(\text{Black} \mid \text{bitten}) = \frac{P(\text{Black and bitten})}{P(\text{bitten})}$$
$$= \frac{\frac{1}{3} \times \frac{1}{2}}{\frac{13}{60}} \quad \checkmark$$
$$= \frac{10}{13} \quad \checkmark$$

Question 27 (2 marks)

Edward plays a game in which he has a probability p of winning, probability q of losing, and probability r of moving to the next round ($p + q + r = 1$).

What is his probability of eventually winning, in terms of p and q ?

2

Let $W =$ 'win round 1', $L =$ 'lose round 1',

$C =$ 'continue after round 1'.

$$\begin{aligned} \text{The } P(\text{eventually wins}) &= P(\text{eventually wins} | W) \cdot P(W) \\ &\quad + P(\text{eventually wins} | L) \cdot P(L) \\ &\quad + P(\text{eventually wins} | C) \cdot P(C) \\ &= 1 \times p + 0 \times q + r \times p(\text{eventually wins}), \end{aligned}$$

since $p(\text{eventually wins} | C) = p(\text{eventually wins})$.

$$\therefore P(\text{eventually wins}) = p + r \times p(\text{eventually wins})$$

$$\therefore p(\text{eventually wins}) = \frac{p}{1-r}$$

$$= \frac{p}{p+q}$$



Question 28 (4 marks)

The point $A(6, 1)$ lies on $h(x)$. The tangent at A is $y = \frac{x}{6}$. Point B is the image of A on the function $g(x) = 3h(2x + 4)$.

(i) Show that B has coordinates $(1, 3)$.

1

$$g(x) = 3h(2x+4) = 3h[2(x+2)]$$

$\therefore$ for any point (x, y) on $h(x)$, the image on

$g(x)$ is $(\frac{x}{2} - 2, 3y)$

$$\therefore B = (\frac{6}{2} - 2, 1 \times 3) = (1, 3)$$



(ii) Hence find the equation of the tangent to $g(x)$ at B.

3

Method 1: tangent to $h(x)$ at A is $y = \frac{x}{6}$

$\therefore$ gradient of tangent to $h(x)$ at A is $\frac{1}{6}$

Thus gradient of tangent to $g(x)$ at B is

$m = \frac{1}{6}$: vertically dilated by factor of 3 &

horizontally dilated by factor of $\frac{1}{2}$

$$\therefore m_B = \frac{1}{6} \times \frac{3}{\frac{1}{2}} = 1 \quad \checkmark \checkmark$$

$\therefore$ tangent to $g(x)$ at B is $y - 3 = 1(x - 1)$

i.e. $y = x + 2$ (or $x - y + 2 = 0$) } $\checkmark$

Method 2:

$$g(x) = 3h(2x+4)$$

$$\therefore g'(x) = 3h'(2x+4) \times 2 = 6h'(2x+4)$$

Since $g'(1) =$ gradient of tangent at B,

$$\therefore g'(1) = 6h'[2(1)+4]$$

$$= 6h'(6)$$

$$= 6 \times \text{gradient of tangent to } h(x) \text{ at A}$$

$$= 6 \times \frac{1}{6}$$

$$= 1$$

$\therefore$ tangent to $g(x)$ at B is $y - 3 = 1(x - 1)$

i.e. $y = x + 2$.

Question 29 (5 marks)

Consider the function $f(x) = \frac{\ln x}{x}$ for $x > 0$.

- (i) Show that the graph of $y = f(x)$ has a stationary point at $x = e$.

2

$$y' = \frac{x \cdot \frac{1}{x} - \ln x \cdot 1}{x^2} = \frac{1 - \ln x}{x^2}$$

Stationary point means $y' = 0 \therefore 1 - \ln x = 0$

$$\therefore \ln x = 1 \therefore x = e$$

So $\therefore$ stationary point at $x = e$.

- (ii) By considering the gradient on either side of $x = e$, or otherwise, show that the stationary point at $x = e$ is a maximum.

1

x	2.71	e	2.72
y'	0.0004	0	-0.00009

$\therefore$ max. at $x = e$.

$$[\text{alt. : } y'' = \frac{(x^2)(-\frac{1}{x}) - (1 - \ln x)(2x)}{x^4} = \frac{-3x + 2x \ln x}{x^4}]$$

$$\therefore y''(e) = -0.0498 < 0 \therefore \text{max at } x = e$$

(iii) Hence deduce that $e^x \geq x^e$ for all $x > 0$.

2

From (ii), $\frac{\ln x}{x}$ ($x > 0$) has a maximum at $x = e$.

$$f(e) = \frac{\ln e}{e} = \frac{1}{e}$$

$$\text{Thus } \frac{\log_e x}{x} \leq \frac{1}{e}, \quad x > 0$$

$$\therefore \log_e x \leq \frac{x}{e} \quad (\text{as } x > 0)$$

$$e \log_e x \leq x$$

$$\therefore \log_e x^e \leq x$$

$$\therefore e^{\log_e x^e} \leq e^x$$

$$\therefore x^e \leq e^x \quad (\text{as } e^{\log_e a} = a)$$

$$\text{i.e. } e^x \geq x^e$$



Question 30 (7 marks)

A truck is making a 1000 kilometre trip at a constant speed of v km/h.

When travelling at v km/h, the truck uses fuel at a rate of $(6 + \frac{v^2}{50})$ litres per hour.

The truck company pays \$2.00 per litre for fuel and pays each of the two drivers \$35 per hour while the truck is travelling.

- (i) Let the total cost of fuel and the driver's pay for the trip be C dollars.

Show that $C = \frac{82000}{v} + 40v$

3

$$\text{Time for trip} = \frac{1000}{v} \text{ hours}$$

$$\therefore \text{Driver cost} = 2 \times \$35 \times \frac{1000}{v} = \$ \frac{70000}{v}$$

$$\begin{aligned} \text{Fuel cost} &= \$2 \times \left(6 + \frac{v^2}{50}\right) \times \frac{1000}{v} = \frac{2000}{v} \left(6 + \frac{v^2}{50}\right) \\ &= \frac{12000}{v} + 40v \end{aligned}$$

$$\therefore C = \frac{70000}{v} + \frac{12000}{v} + 40v$$

$$\therefore C = \frac{82000}{v} + 40v \text{ dollars.}$$

- (ii) The truck must take no longer than 12 hours to complete the trip, and speed limits require that $v \leq 110$.

At what speed v should the truck travel to minimise the cost C ?

4

(you may disregard any change-over time for the drivers to swap).

$$C = 82000 v^{-1} + 40v$$

$$\therefore \frac{dC}{dv} = -\frac{82000}{v^2} + 40$$

$$\text{For min, } \frac{dC}{dv} = 0 \therefore \frac{82000}{v^2} = 40 \therefore v^2 = 2050$$

$$\therefore v = \sqrt{2050} \approx 45.3 \text{ km/h.}$$

But, if $v = \sqrt{2050}$, $t = \frac{1000}{\sqrt{2050}} \div 22.1$ hours

This is too long, as we need $t \leq 12$ (ie. $v \geq 83\frac{1}{3}$)

In fact, the domain here is $83\frac{1}{3} \leq v \leq 110$.

So we need to check the cost at the endpoints of the domain.

$$\text{If } v = 83\frac{1}{3}, C = \frac{82000}{83\frac{1}{3}} + 40(83\frac{1}{3}) \div \$4317$$

$$\text{If } v = 110, C = \frac{82000}{110} + 40(110) \div \$5145$$

So we get minimum cost when $v = 83\frac{1}{3}$ km/h.

[Since $\frac{d^2C}{dv^2} = 2 \times 82000 v^{-3} > 0$ for all $v > 0$

So $v = \sqrt{2050}$ gives min. turning point &

C is increasing after $v = \sqrt{2050}$. As v increases,

C increases, so the minimum C that satisfies the

conditions corresponds to $v = 83\frac{1}{3}$ km/h.

But all that's required is to calculate C at the endpoints.]

ENDS